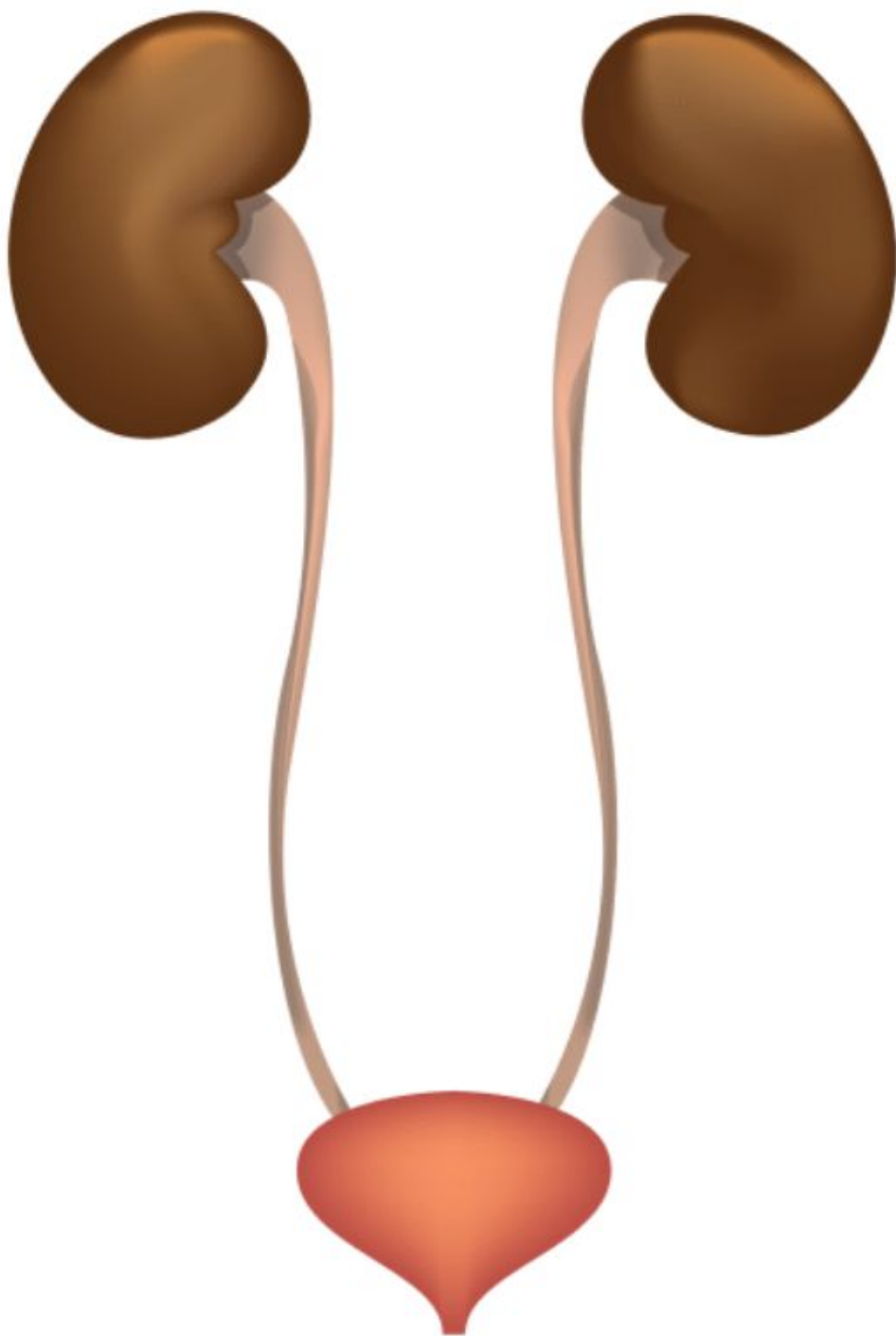


PLABABLE

GEMS 

VERSION 1.1

UROLOGY



Ureteric Stones

Presentation

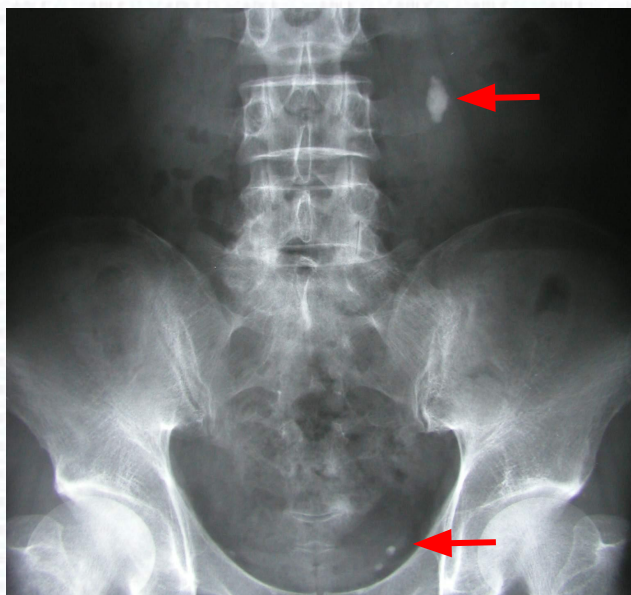
- Loin to groin acute severe pain
- Haematuria
- Dysuria
- Nausea and vomiting

Risk factors

- Anatomical abnormalities such as ureteric stricture or horseshoe kidneys
- Gout
- Hyperparathyroidism
- Dehydration
- Cystinuria

Investigation

- **Non-contrast CT scan (Investigation of choice)**
- Ultrasound for radiolucent stones



Ureteric Stones

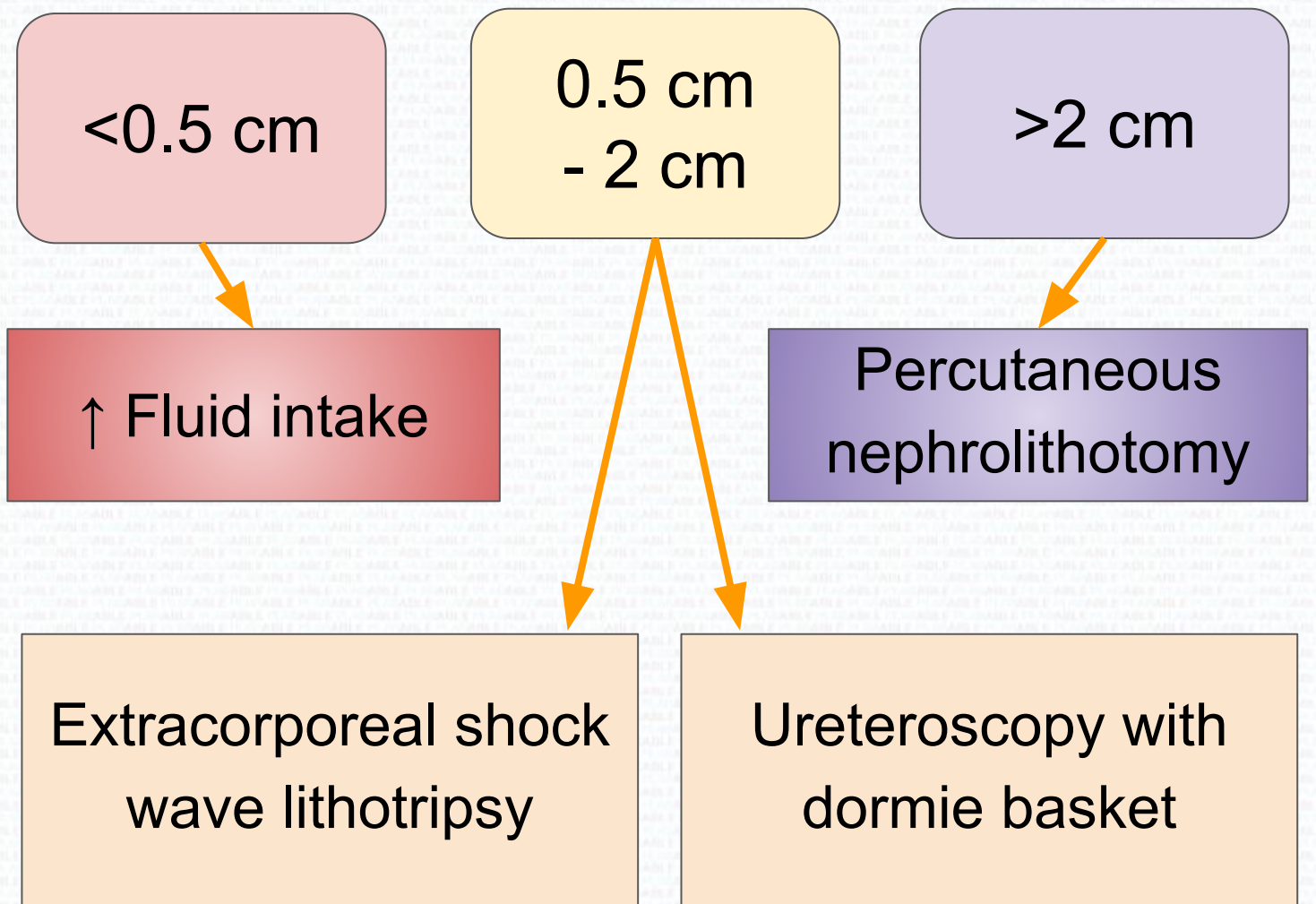
Management	
Symptomatic stones < 10 mm	SWL (first line) URS (second line)
Symptomatic stones 10 - 20 mm	URS or SWL
Symptomatic stones > 20 mm	PCNL
Asymptomatic stones < 5 mm	Watchful waiting
Asymptomatic stones > 5 mm	Explain the risk and benefits Wait and watch if the patient agrees

*SWL - Shock wave lithotripsy
URS - Ureteroscopy
PCNL - Percutaneous nephrolithotomy

Ureteric Stones

- Provide painkiller (NSAIDs) before definitive management
- Stones < 5 mm usually pass on their own with adequate hydration
- Patient with only one kidney plus hydronephrosis → **percutaneous nephrostomy** (decompression)
- Patient with two kidneys +AKI and hydronephrosis → **percutaneous nephrostomy**

Managing Renal Stones



Percutaneous nephrostomy vs. nephrolithotomy

Stoma catheters to pelvicalyceal system of kidney for decompression.
→ Draining the obstructed fluid in kidney

Removal of urinary stone percutaneously via scope (if size >2cm)

Managing Renal Stones Complications

Obstructive uropathy:

With 1 functioning kidney:

- A stone (**any size**) with dilation of pelvicalyceal system (PCS)
- ± anuria
- ± fever

→ Treat with **decompressing PCS** by percutaneous nephrostomy

With 2 kidneys:

- AKI
- Fever
- Hydronephrosis

→ Treat with **decompressing renal collecting system** by percutaneous nephrostomy

Managing Renal Stones

Complications

Summary

- Loin pain + stone + hydronephrosis
→ Manage according stone size
- Loin pain + stone + hydronephrosis + **AKI + Fever**
→ Percutaneous nephrostomy

Scrotal Swelling

Condition	Features
Inguinal hernia	● Inguinoscrotal swelling ● Not able to get above the swelling ● Cough impulse + ● Reducible
Hydrocoele	● Soft fluctuant swelling ● Painless and non-tender ● Able to get above the swelling ● Transillumination +
Epididymal cyst	● Single or multiple slow growing cysts ● Painless and non-tender ● Lies above and behind the testes ● Able to get above the swelling
Acute epididymo-orchitis	● Dysuria ● Urethral discharge ● Fever ● Red and tender scrotal skin ● Pain reduce on elevation of the testis

Scrotal Swelling

Condition	Features
Testicular tumour	● Discrete testicular nodule ● Firm to hard in consistency ● USG scrotum ● Serum AFP and β-HCG
Varicocele	● Dilation of the pampiniform plexus ● Common in the left side ● Dull aching or dragging pain ● Swelling $\uparrow$ on standing ● Swelling $\downarrow$ on lying down ● Bag of worm appearance ● Scrotal doppler
Testicular torsion	● Acute, severe testicular pain ● Common in adolescent and young ● Pain does not reduce by elevation of the testis ● Urgent exploratory surgery with Orchidopexy

Vesicovaginal Fistula

Presentation: Continuous and involuntary urine discharge

Pathology: Connection between bladder and vagina

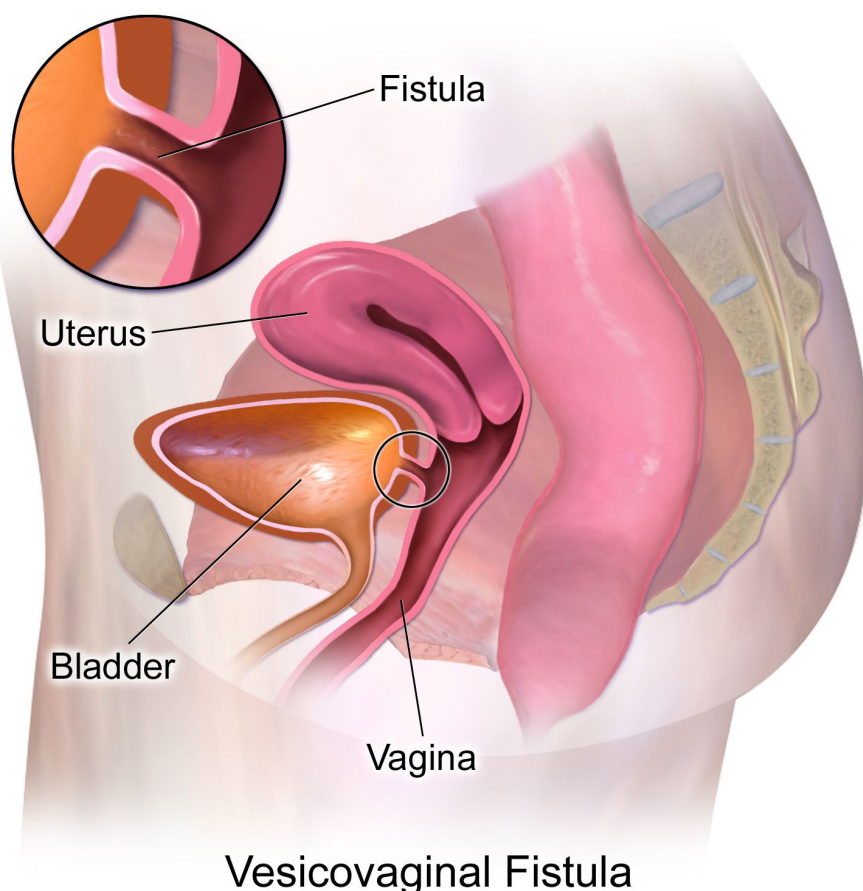
Causes:

- Obstetric fistula (prolonged labor)
- Hysterectomy
- Radiation therapy

Investigation

- 3 swab test
- USG

Management: Surgical treatment



Urinary Incontinence

Stress incontinence - incompetent sphincter:

- Leakage of urine during activities which ↑ intraabdominal pressure such as sneezing, coughing or laughing
- **Cause:** Multiple vaginal delivery (pelvic floor muscles become weak)
- **Treatment:**
 - Pelvic floor exercises (first-line)
 - Surgical - open colposuspension

Urge incontinence - detrusor overactivity:

- Difficulty in controlling the desire to urinate
- Wetting before making it to the bathroom
- **Treatment:**
 - First → **Bladder training** - gradually ↑ the period between voiding
 - Second → Anticholinergic drugs: e.g. **oxybutynin** (only after trying bladder training)

Urinary Incontinence

Overflow incontinence - bladder outlet obstruction:

- Continuous overflow and difficulty in completely emptying the bladder
- **Causes**
 - BPH
 - Prostate cancer
- **Treatment**
 - Specific for BPH or prostate cancer
 - Intermittent catheterisation

Benign Prostatic Hyperplasia

Presentation

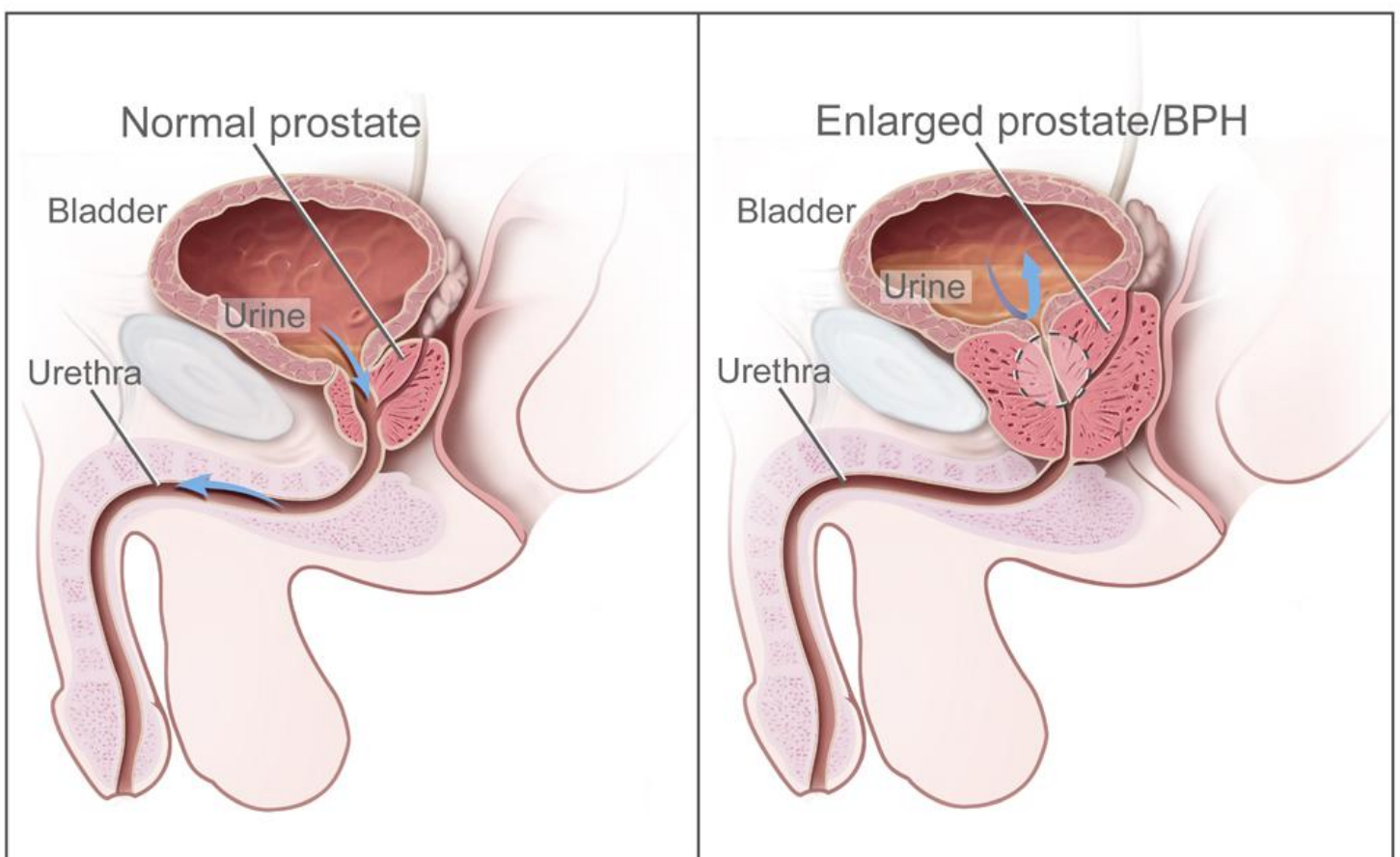
- ↑ Urinary frequency
- Incomplete bladder emptying
- Poor stream and dribbling of urine
- Hesitancy
- Common in elderly men

Digital rectal examination: shows smooth and enlarged prostate

Note: Hard and nodular prostate indicates prostate cancer

Investigations

- ↑ PSA
- ↑ Post-void residual bladder volume



Benign Prostatic Hyperplasia

Treatment

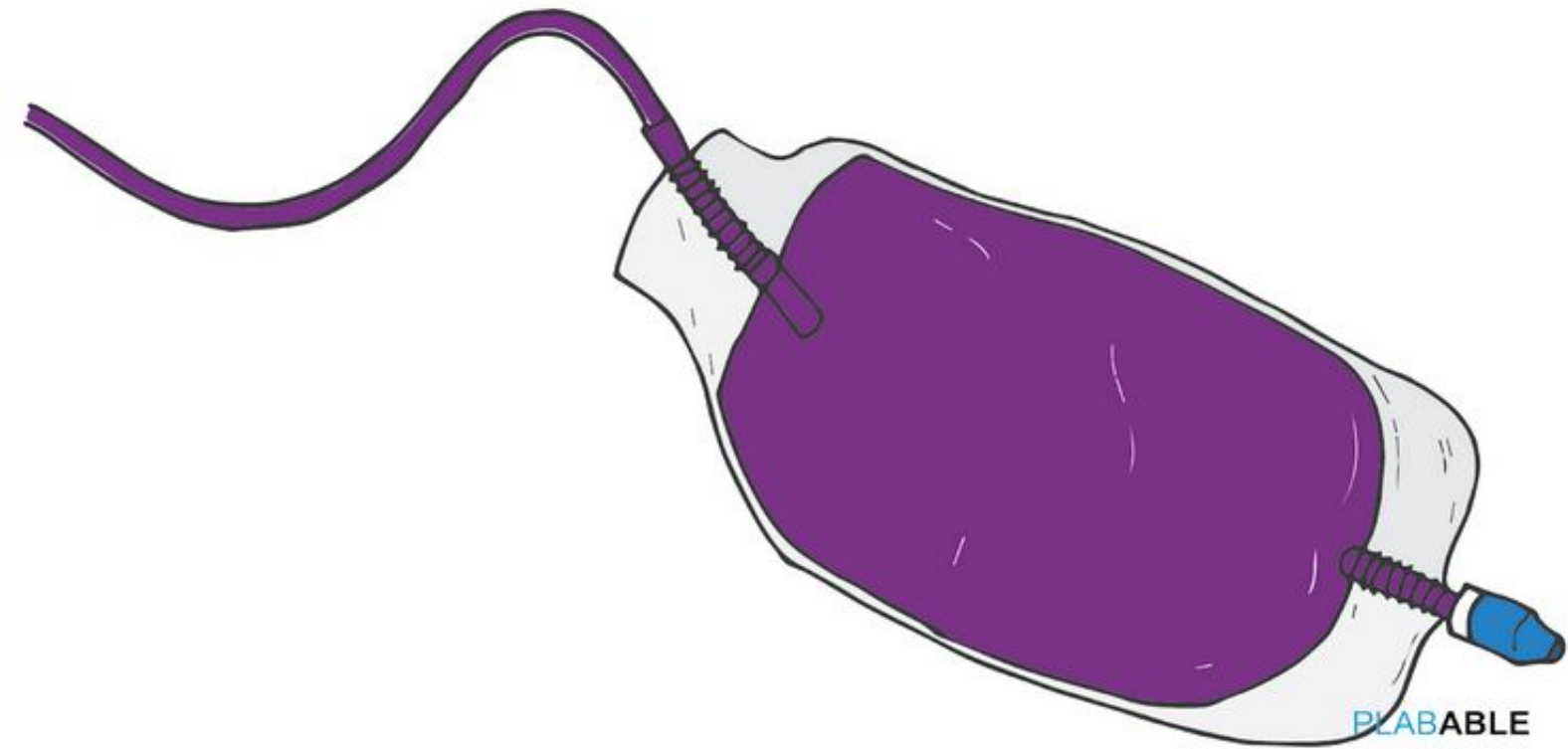
- **Selective α -1a blockers:**
 - Tamsulosin
- **Less selective α -blockers** (if the patient also has hypertension to treat):
 - Prazosin
 - Doxazosin
 - Terazosin
- **5- α reductase inhibitor**
 - Finasteride
 - Dutasteride
- **Transurethral resection of prostate - TURP**
(If medical management fails)

α -1 blockers - Mild to moderate lower urinary tract symptoms alone

5- α reductase inhibitor - Urinary tract symptoms plus significant prostatomegaly or PSA>1.4 ng/mL

TURP syndrome can cause hyponatremia due to irrigation of bladder used during the procedure

Purple Urine Bag Syndrome



Brain trainer:

For the above pictured condition, what is the etiology indicated investigation and management?

- ➔ Bacterial colonization of the urinary tract
- ➔ Urine culture
- ➔ Antibiotics and changing of the catheter

Prostate Cancer

Features

- Most common cancer in men
- Sensation of incomplete emptying
- ↑ Urinary frequency and urgency
- Haematuria
- Weight loss and back pain (metastasis)
- Metastasis - can lead to Cauda Equina syndrome presenting with saddle anesthesia & inability to urinate (urgent MRI spine)

Digital rectal examination

- Asymmetrical, irregular, hard & enlarged prostate

Investigations

- ↑ PSA (initial investigation)
- Prostate biopsy (definitive)
- MRI scan

Management

- **Localised** - radical prostatectomy & radiotherapy
- **Metastatic** - radical radiotherapy & androgen deprivation treatment

Bladder Cancer

Presentation

- Painless haematuria
- Sometimes voiding symptoms
- Common type: transitional cell carcinoma

Risk factors

- Smoking
- Occupational exposure to **aniline** dye

Investigation

- Cystoscopic examination of bladder and biopsy
- Urine cytology

Note: First differential for an **elderly male** with haematuria should be bladder cancer, and for an **elderly female** with vaginal bleeding should be endometrial cancer

Testicular Cancer

Presentation

- Painless and non-tender lump in the body of testes
- Testicular pain
- Metastasis - bone pain
- > 95% are germ cell tumours

Risk factors

- Cryptorchidism
- Klinefelter syndrome

Investigations

- Scrotal ultrasound
- AFP - Yolk sac tumour
- Beta-hCG - Both teratoma and seminoma

Note: Testicular biopsy is not done as it can cause metastasis

Management

- Radical orchiectomy
- Chemotherapy (if metastasis)

Testicular Torsion

Rotation of the testes through the spermatic cord
thereby occluding testicular blood vessels

Presentation

- Sudden and severe pain in one testes
- Happens during sports or activity
- Lifting the testes does not relieve the pain (reduces the pain in epididymo-orchitis)
- Absent cremasteric reflex

Investigation

- Colour doppler USG to look for blood flow

Management

- Emergency surgery and testicular detorsion followed by bilateral orchidopexy

Epididymo-Orchitis

Presentation

- Unilateral scrotal pain and swelling
- Urethral discharge
- Tenderness to palpation of epididymis
- Fever
- History of UTI
- Pain relieved by elevation of the testes
- <35 years old sexually transmitted organisms:
 - *Chlamydia* & *Gonorrhoea*
- >35 years old UTI causing organisms:
 - *E. coli* & *Pseudomonas. spp*

Investigation

- Urethral swab and smear
- Microscopy and culture of mid-stream urine

Treatment

- **Sexually transmitted pathogen:**
 - Ceftriaxone 250 mg IM Stat PLUS
 - Doxycycline 100 mg BD for 10-14 days
- **Enteric organisms:**
 - Ofloxacin 200 mg BD for 14 days OR
 - Ciprofloxacin 500 mg BD for 10 days

AD - Polycystic Kidney Disease

Presentation

- Loin pain
- Haematuria
- Hypertension

Associated with intracranial (Berry) aneurysm

Investigations

- Abdominal USG (initial test)
- MRI / CT (most sensitive)

Management

- Hypertension with ACEi or ARB
- Regular screening for intracranial aneurysm



Haematuria DDs

Goodpasture syndrome	● Haematuria ● Hemoptysis ● Anti-GBM antibodies
Wegener's granulomatosis (granulomatosis with polyangiitis)	● Haematuria ● Haemoptysis ● Arthritis
Haemolytic uremic syndrome	● Haematuria ● Bloody diarrhoea ● Haemolytic anaemia ● Thrombocytopenia ● Infection with E.coli
Bladder cancer	● Haematuria ● Occupational exposure - Aniline dye ● Old age
Post-streptococcal glomerulonephritis	● Haematuria ● H/o Impetigo or pharyngitis ● Oedema ● Hypertension ● Children

Vesicoureteric Reflux

Features

- Urine flows retrograde from bladder into the ureter and kidney
- Causes dilation of pelvicalyceal system
- Repeated UTIs
- Reflux nephropathy (if chronic)

Investigations

- **Initial:**
 - Ultrasound
 - Urinalysis and C&S
- **Gold standard:**
 - Micturating cystourethrogram
- **Detect parenchymal scars:**
 - DMSA scan

Management

- Prophylactic antibiotics to prevent recurrent UTIs
- Surgical for high grade reflux



Grade V
VUR

Interstitial Cystitis

Presentation

- ↑ Urinary frequency
- Urgency
- Suprapubic pain ↑ with bladder filling and ↓ with voiding
- Diagnosis of exclusion
- Common in women

Investigations

- Urine routine and C&S to R/O UTI
- Cystoscopy to R/O bladder cancer

Management

- Behavior therapy
- Avoiding food triggers
- Analgesics
- Amitriptyline

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